BEFORE YOU READ

### Subspaces built from two subspaces

From two subspaces $W_1$ and $W_2$ of $V$, the intersection and the sum are again subspaces, and they bracket each of the originals.

$$W_1 \cap W_2 \subseteq W_1 \subseteq W_1 + W_2 \subseteq V$$

$V$ **The whole space**

$W_1 + W_2$ **Sum**

adds all combinations · the smallest subspace containing both

$W_1$ **Subspace**

one of the two originals

$W_1 \cap W_2$ **Intersection**

always a subspace; between {0} and one of the originals

TAKEAWAY

### Dimension bookkeeping

$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$. The union, unlike these, is almost never a subspace.